

Validator Anonymity

The P2P network still has a privacy issue

Yann Vonlanthen · Ethereum Foundation

Who runs which validator?

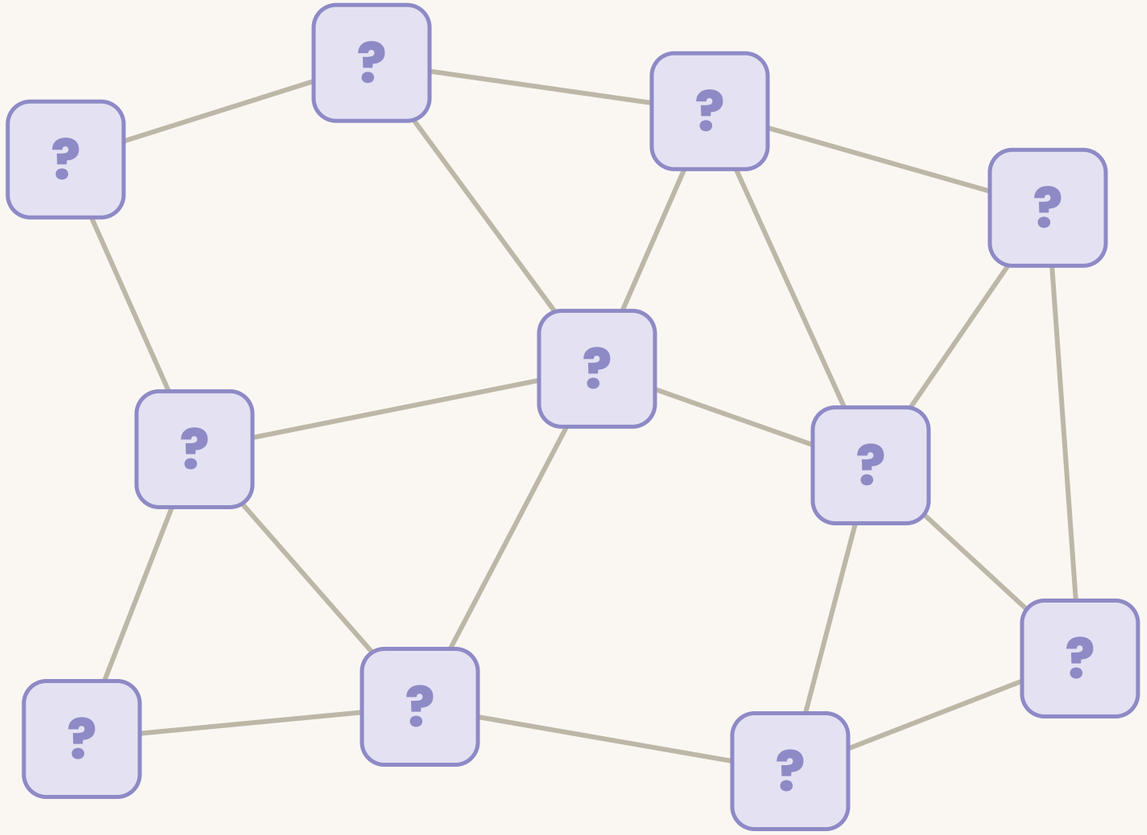
On chain

validator 93

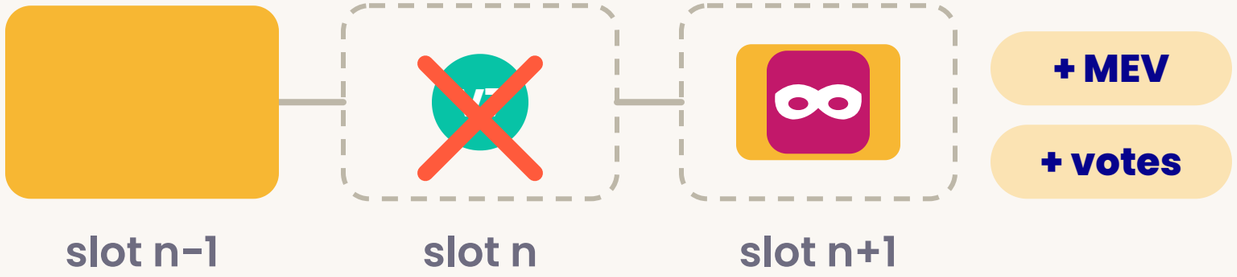
?

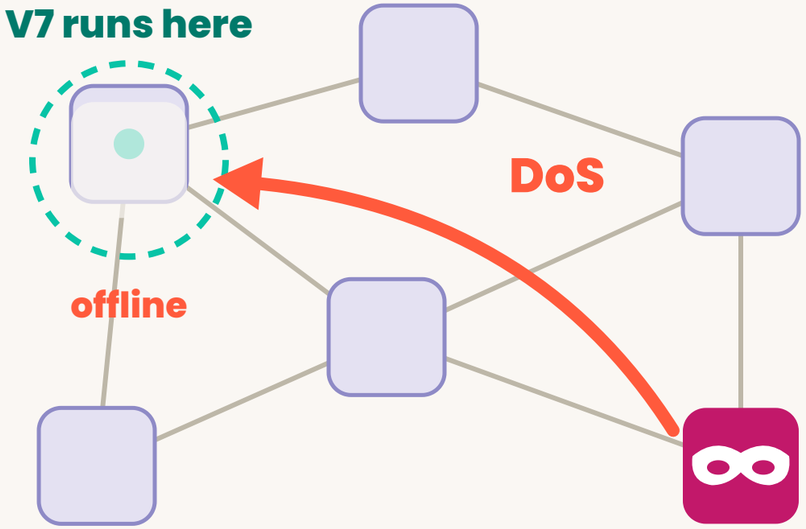
On the network

203.0.113.7



Why does anonymity matter?



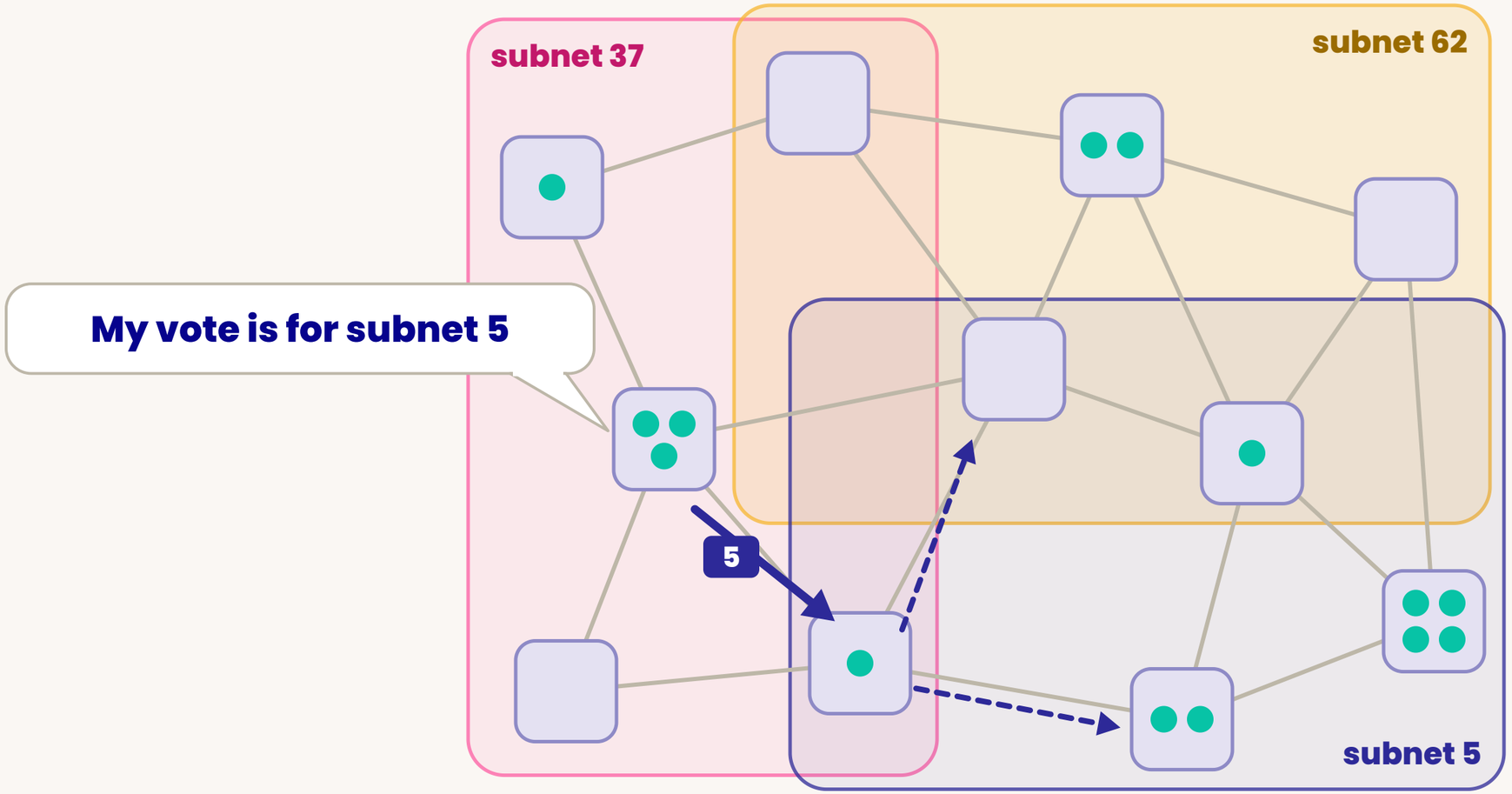


Our 2024 talk:

“The proposer of slot n+1 could snipe the rewards of the proposer of slot n.”

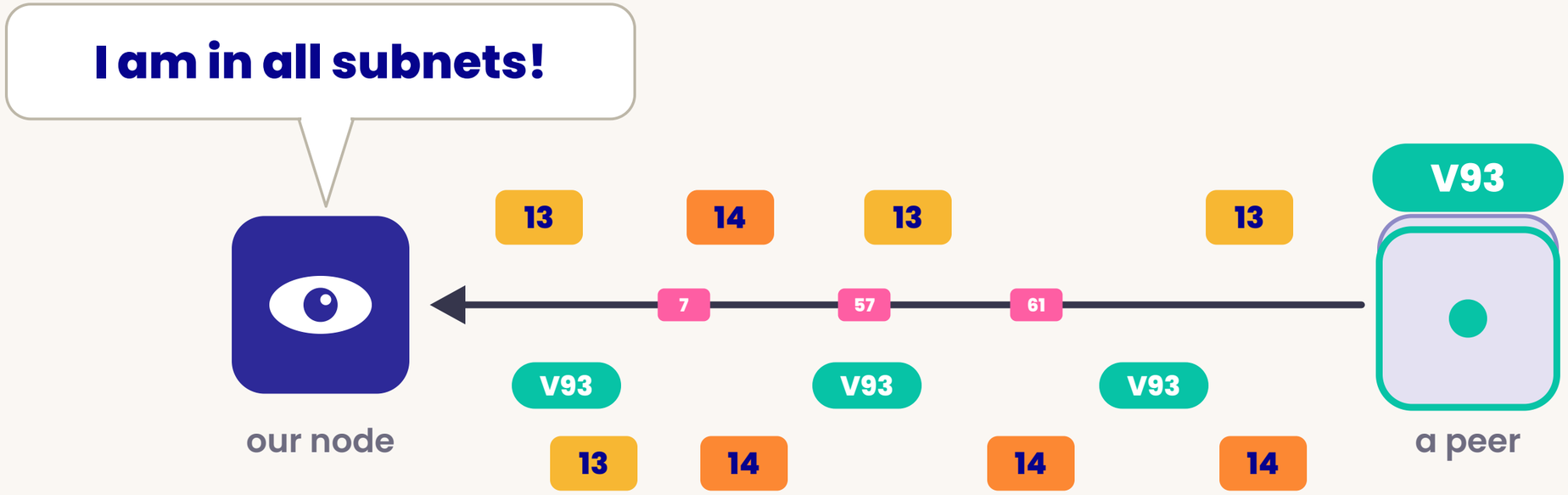
“Finality might be impacted.”

64 subnets



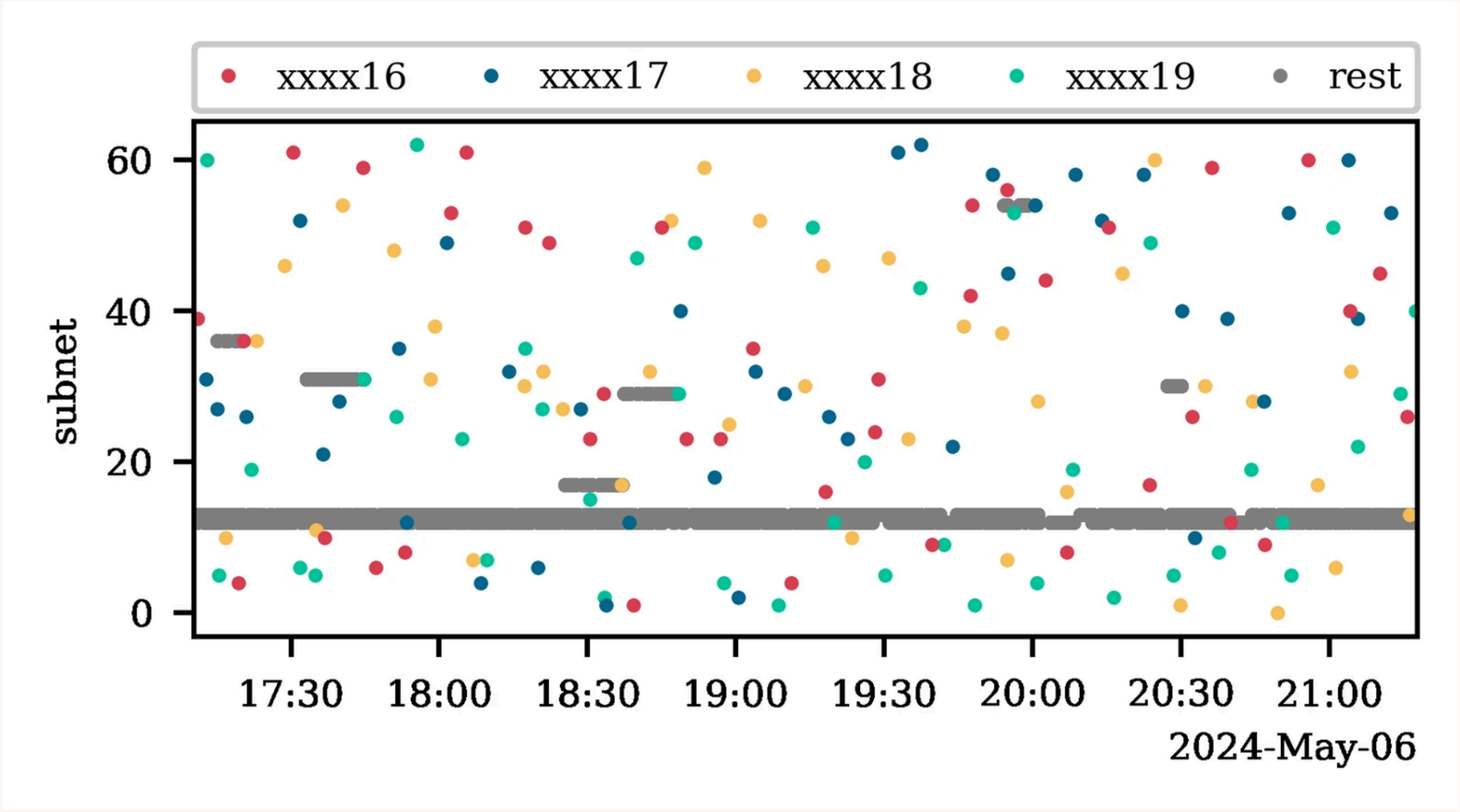
Do we give up privacy for lower network load?

Listening to everything



Outside its two subnets, it only sends its own validator's votes

One peer, one evening



Every message from one peer

The band: its two subnets

Red: one validator

once per epoch, any subnet

Four of them. All on this peer.

4 nodes, 3 days

Frankfurt · Seoul · Virginia · Zurich

161,057

validators located · more than 15 % of all

EF bug bounty

May 2024 · Heimbach, Vonlanthen, Villacis, Kiffer, Wattenhofer, USENIX Security 2025

Few machines

19,390

validators on one machine

a large staking pool

A few hundred machines could stall finality.

Few clouds

~90 %

of located validators run in the cloud

Amazon alone: 19 %

2024

An attacker could

- snipe a proposer’s rewards
- stall finality

2026

With numbers

- KnockBlock: +44.5 % MEV
- StakeBleed: ~300 ETH in 2 h

Doumanidis & Apostolaki, Princeton
Financial Cryptography 2026

Find, hijack, exploit

Routing Attacks in Ethereum PoS · Doumanidis & Apostolaki · FC 2026

1

Find

Every validator,
to its IP prefix

2

Hijack

Steal the route
to that prefix

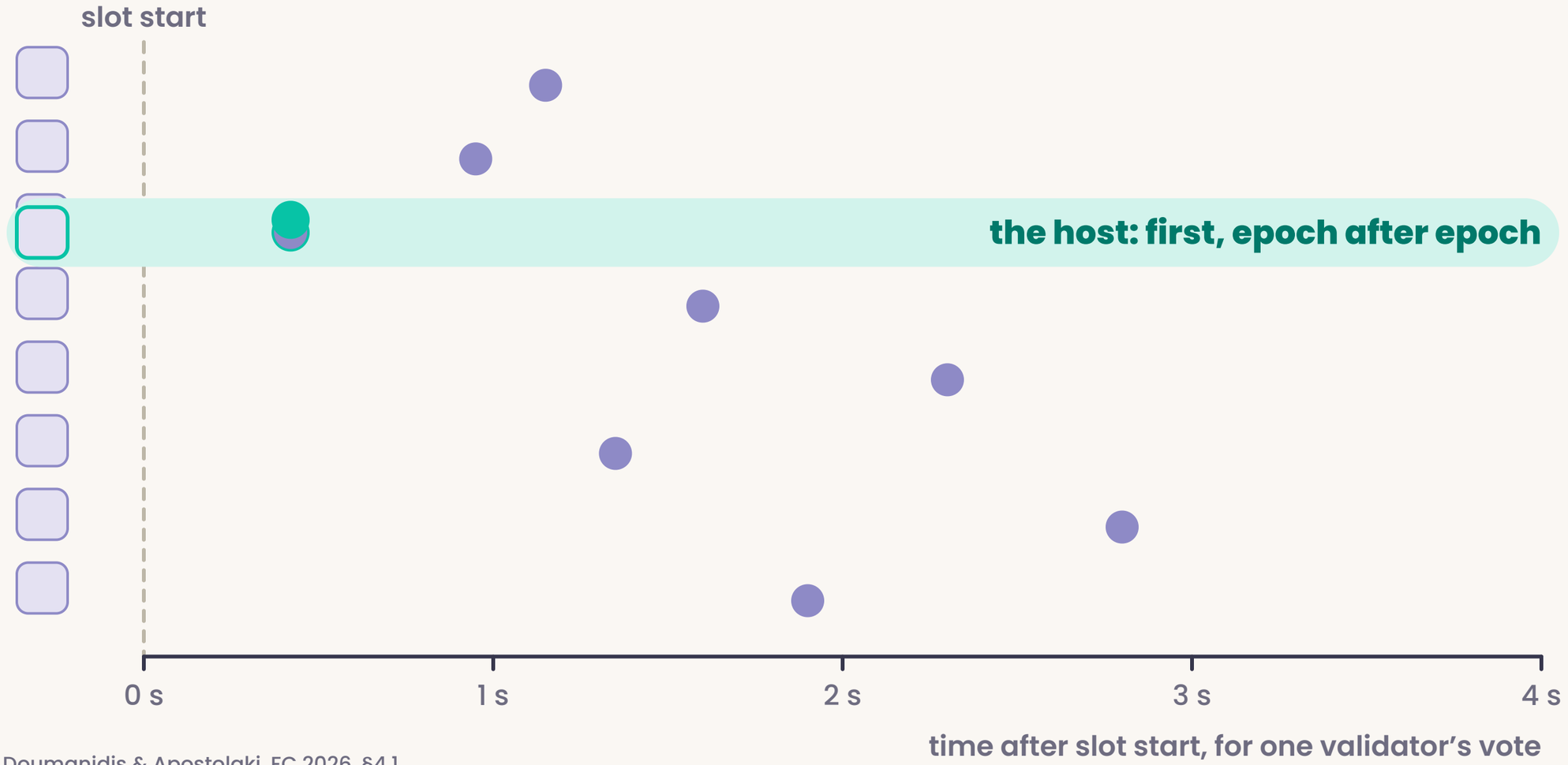
3

Exploit

StakeBleed
KnockBlock

The host speaks first

Their SuperNode times every vote that is pushed to it eagerly



Schematic · Doumanidis & Apostolaki, FC 2026, §4.1

From a seed to everyone

Seed

consecutive IDs
share a prefix
(50 % of validators)

Clean

same deposit key
same hosting AS
for training

Learn

timing classifier
91 % accurate

Map

1,063,660

validators, to
4,600 prefixes

Data: Nov 2024 to Jan 2025 · Doumanidis & Apostolaki, FC 2026, §4.2

From 15 % to everyone

Joining all 64 subnets no longer hides you.

Where both studies overlap with high confidence, they agree on 70 % · Doumanidis & Apostolaki, FC 2026

Validators live in few places

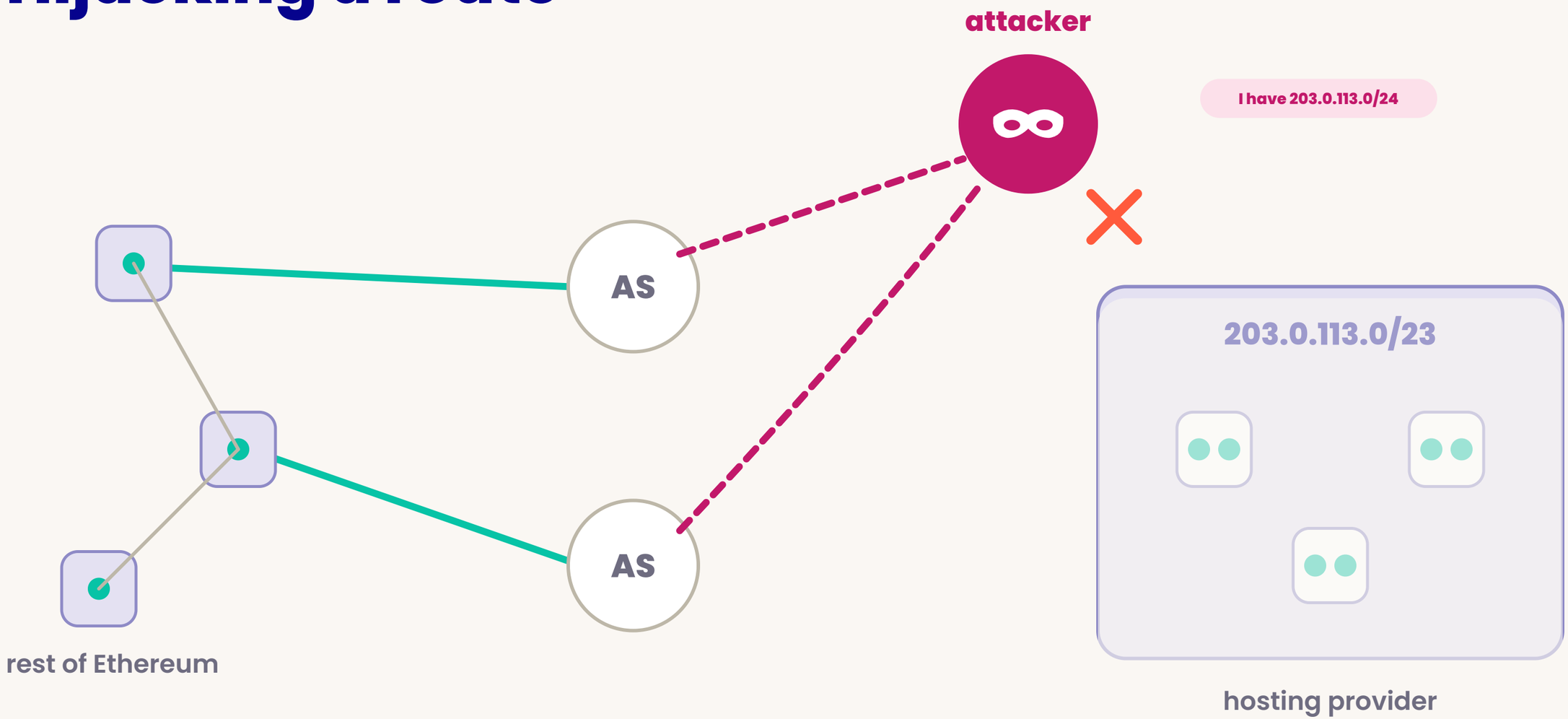
4,600 prefixes host all validators

29

prefixes hold a third of all validators

Three hosting networks hold more than half · OVH alone ~27 %

Hijacking a route



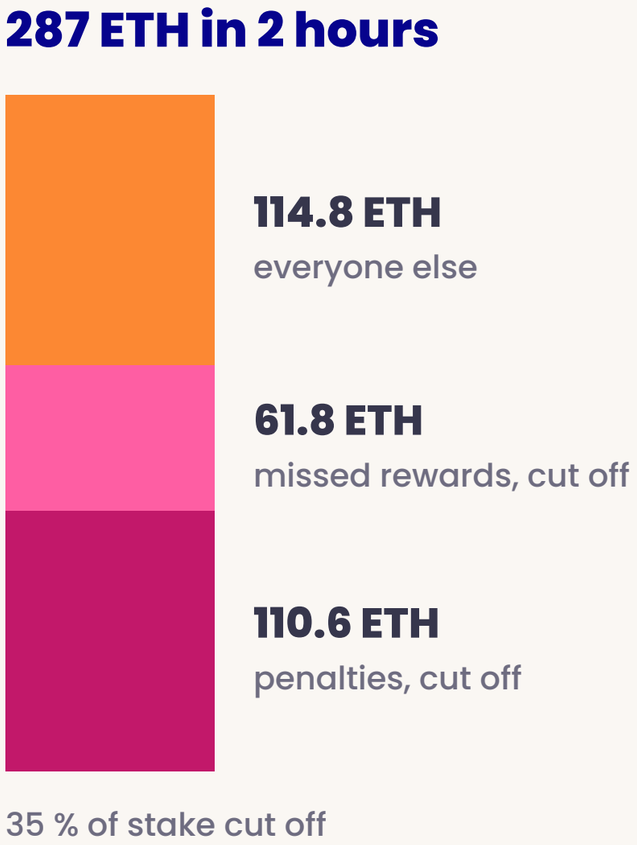
- 40.85 % of Ethereum nodes: no proper RPKI protection
- Undoing a hijack: 2 to 3 hours, by hand

StakeBleed



More than 1/3 of the stake is cut off

40 % of the loss hits validators that were never cut off



Doumanidis & Apostolaki, FC 2026, §3.2 and §6 · 1,063,660 validators · even without a leak, 20 % cut off for 2 h costs 214 ETH

KnockBlock



< 2 min

one prefix hijacked, for one slot

+44.5 % MEV

on average, in the block after a missed slot

Pays off from 4 validators.

2024: could · 2026: +44.5 %

Doumanidis & Apostolaki, FC 2026, §3.3 and §7 · break-even assumes ~$2,500 a year to run an AS

Since 2024

new leak

Nodes advertise their stake

Fulu, Dec 2025: ENR field cgc, stake / 32 ETH

cgc	cgc	cgc
4	8	64
no validators	some stake	~2,048 ETH

in state

Proposers fixed 2 epochs ahead

Fulu: lookahead in the state (EIP-7917) · Whisk: stagnant

unchanged

Still 2 of 64 subnets per node

The subnet trick still works

consensus-specs: fulu/validator.md, fulu/p2p-interface.md (cgc), phase0/p2p-interface.md (SUBNETS_PER_NODE) · EIP-7917 Final, EIP-7441 Stagnant

Our 2024 question

Do we give up privacy for lower network load?

Better trade-offs

- Remove the signal
- Spend the bandwidth dividend
- Hide in your own subnet (DC)
- Vote through a throwaway relay
- Keep roles secret
- Make isolation hard

Noise doesn’t hide you

Their mapping model



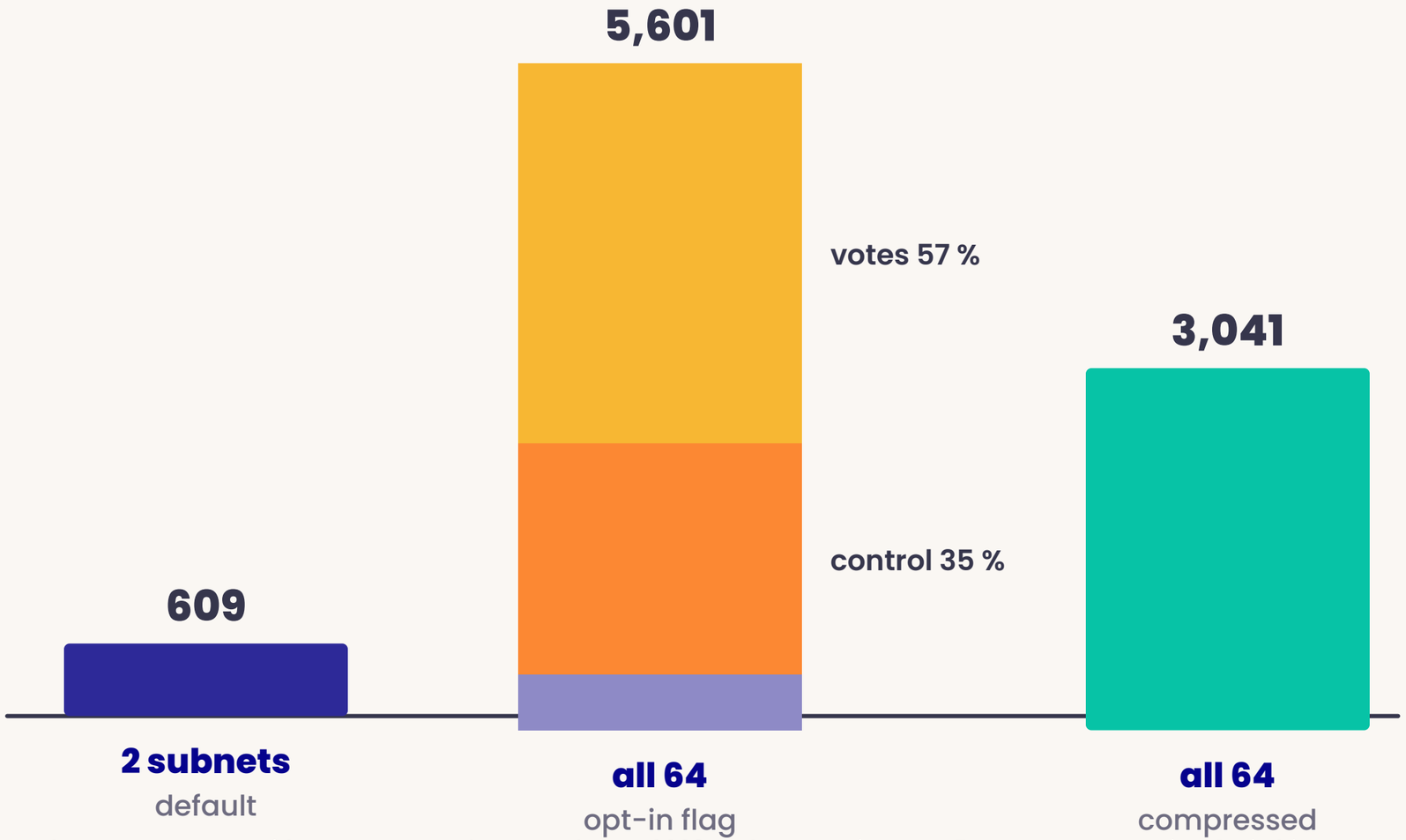
93 %

Remove the signal. Noise is cheap to filter out.

Doumanidis & Apostolaki, FC 2026, §8

Spend the bandwidth dividend

Inbound MiB per hour, one mainnet node



9.2×

to join all 64 subnets

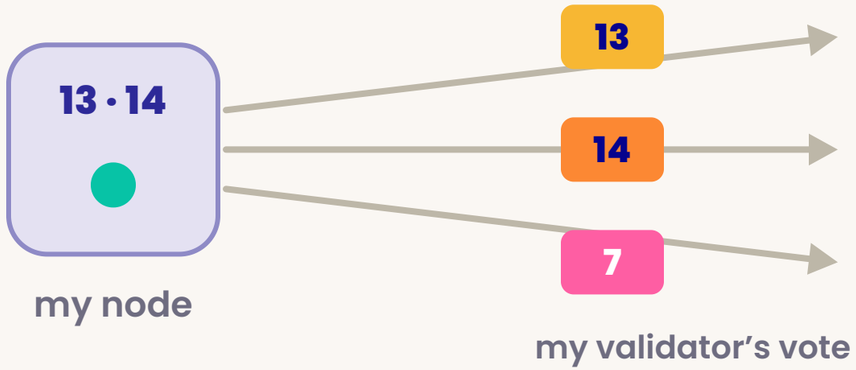
Stream compression:
−46 %

Beats the subnet trick.
Not timing.

Nashatyrev, “Snappy with a memory”, ethresear.ch, Sep 2026 · Teku on mainnet, ~45 peers · compression: 1 MiB history

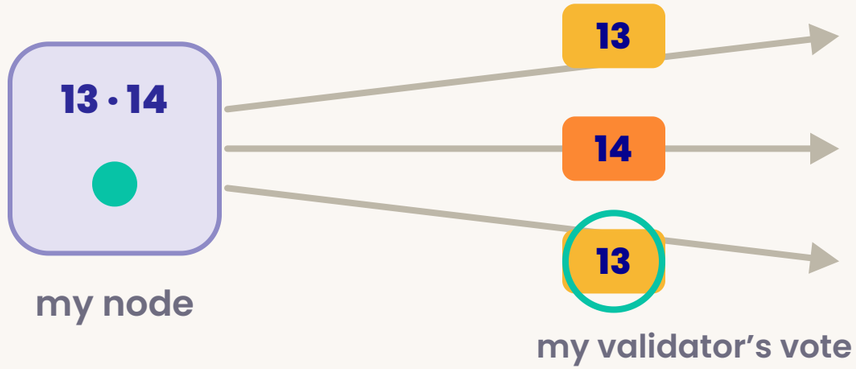
Hide in your own subnet

Today



Stands out

DC proposal



Looks like relaying

Validators keep their committee and subnet. Cost: +1.90 Mbit/s per subnet, vs 0.55 today

Hides from the subnet trick, not from timing

dc-networking: properties.md (anonymity enhancing), staggered-committees/analysis/network.md §4 · model: 1M validators, kappa = 7

Never fill a committee alone

Contiguous indices: one operator fills the committee



Mapped in one observation: exact and free

Strided or era-shuffled: every committee mixed



Same schedule, at most one reshuffle a day

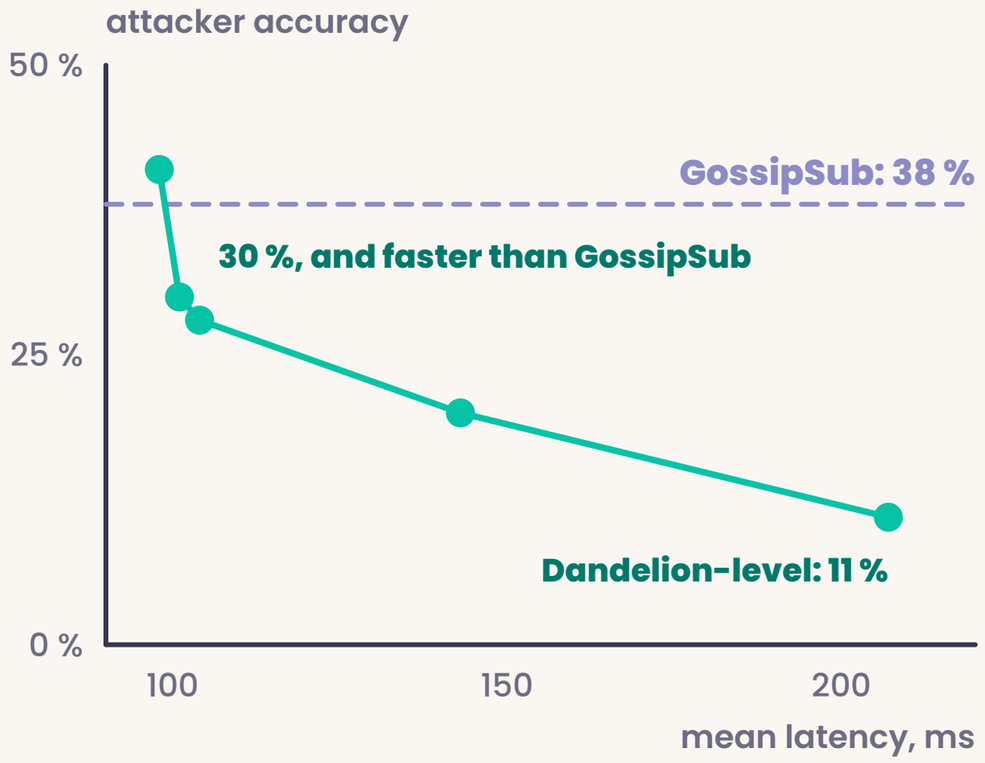
32 to 93 %

of an operator's validators
in committees it fills alone

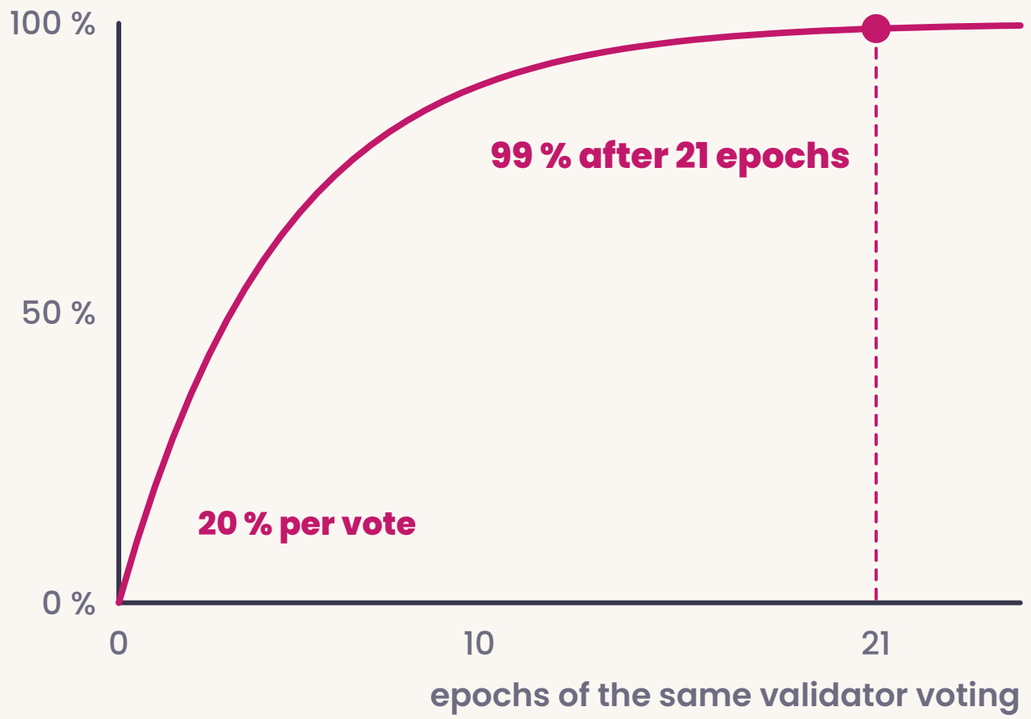
dc-networking: staggered-committees/analysis/network.md · deposit batches of 1,000 to 10,000 validators

Hide each message?

SPREAD, per message



But votes repeat every epoch

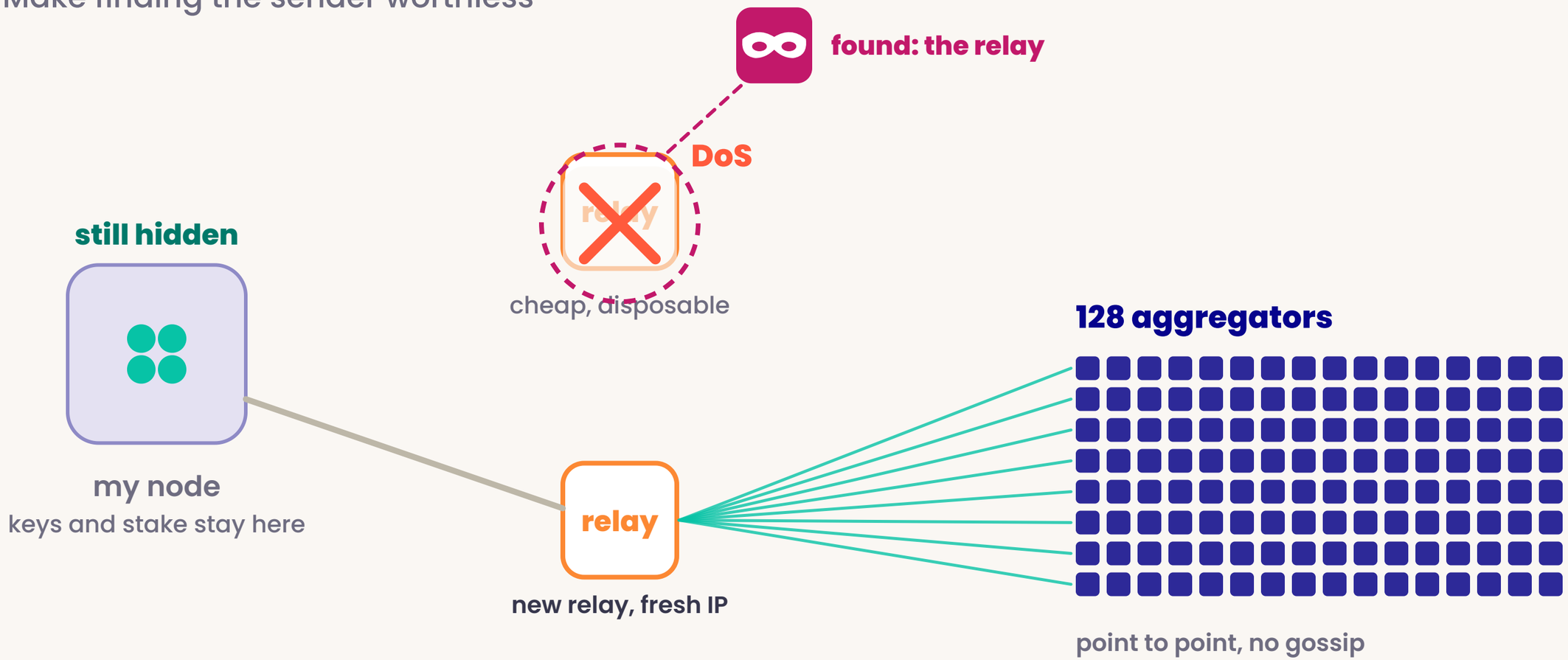


Worth it for one-shot messages: blocks, inclusion lists. Not for votes.

SPREAD, ethresear.ch 25343, Jul 2026: simulation, 10 % curious nodes · right: 5 % curious nodes, 20 % per vote, from a reply in that thread

Vote through a throwaway relay

Make finding the sender worthless



Votes can skip gossip: straight to 128 aggregators, Wonderboom-style

Close to the paper's temporary proxy nodes (Doumanidis & Apostolaki §8) · Wonderboom, arXiv 2602.06655 · dc-networking open question P4

Throwaway relays: the catch

The node still leaks

Since Fulu its ENR shows its stake (cgc): StakeBleed can still find its prefix.

Relays are re-found

A new relay's first votes map it again. Each respawn costs missed votes.

Proposals need more

A known relay can be DoSed in your slot: use a fresh one. Builder API calls leak too.

Aggregators exposed

Direct sending needs known, reachable aggregators, and 128 copies of every vote.

Who gets in?

Fresh relays have no peer score: floods are cheap. Needs PoV or per-vote checks.

Trust and clouds

A shared relay sees, and can drop, your votes. Cheap relays crowd into a few clouds.

"Partial protection is no protection."

ethresear.ch 21050, Nov 2024, on Builder API registrations

Keep roles secret



- Whisk (SSLE): stagnant. Fulu made the lookahead public state
- Fresh validator keys every day: SSLE almost for free
- DC: secret duties via VRF, +96 B per vote

EIP-7441 (Stagnant), EIP-7917 (Final) · Buterin, “The Extremely Lean Chain”, ethresear.ch, Jul 2026 · dc-networking network.md §4

Make isolation hard

- Sign your routes (RPKI)
- Spread across providers
- Detect a hijack, move automatically

Operators can do this today.

40.85 %

of Ethereum nodes lack proper RPKI protection

OVH alone: ~27 % of validators

Three ASes: more than half

Doumanidis & Apostolaki, FC 2026, §3.1, §5, §8

Anonymity is a security property.

- Protocol: stop leaking signals
- DC: own subnet, never alone
- Operators: sign routes, spread out

Thanks!

@yannvon

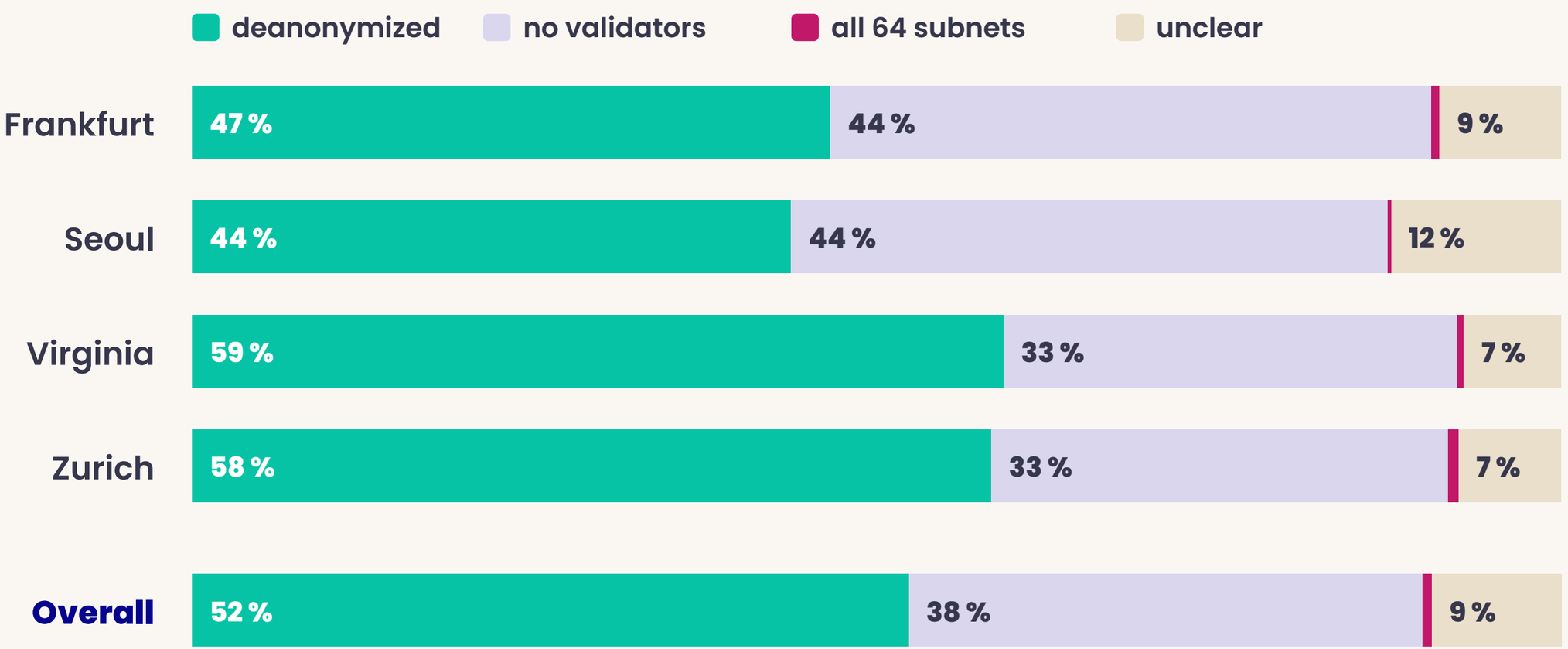
Heimbach, Vonlanthen, Villacis, Kiffer, Wattenhofer · USENIX Security 2025

Doumanidis & Apostolaki · FC 2026

Appendix

For Q&A

Which peers we could read



Peers with long connections (> 32 epochs), 4,372 in total · Heimbach et al., USENIX Security 2025

Our heuristic: v runs on p if

C1 At least 90 % of the expected non-backbone votes of v arrive from peer p

C2 p is not subscribed to all 64 subnets

C3 We receive at least every tenth vote we expect for v from p

C4 v's vote count from p is one standard deviation above p's mean per validator

Conservative on purpose: we only label peers we saw long and well enough.

Heimbach, Vonlanthen, Villacis, Kiffer, Wattenhofer, USENIX Security 2025, §4

Validators per machine

27 %	of peers host a single validator
> 50 %	host at most 4
11 %	host more than 100
709	per machine on average, five largest pools
19,390	on a single machine

Operators split validators in round numbers: 100, 250, 300, 350, 400, 500, 1,000

StakeBleed, in numbers

11,072 Gwei	average reward per validator per epoch, without attack
4 epochs	cut off before the inactivity leak starts, ~26 min
287 ETH	35 % cut off for 2 h: 110.6 penalties + 61.8 missed + 114.8 others
214 ETH	20 % cut off for 2 h, no leak
> 125 ETH	per epoch lost by the connected validators during a leak
~3 weeks	of leak until a 32 ETH validator is ejected at 16 ETH

Doumanidis & Apostolaki, FC 2026, §3.2, §6 · validator set of 1,063,660

KnockBlock, in numbers

12.8 min	proposer schedule known in advance (2 epochs)
< 90 s	for a BGP announcement to take effect
+44.5 %	MEV on average after a missed slot
+453.6 %	proposer reward at most, from the extra votes
0.19 ETH	extra per validator per year
> 3	validators to cover ~$2,500 a year for an AS

Doumanidis & Apostolaki, FC 2026, §3.3, §7

SuperNode, in numbers

Eager only	keeps votes pushed directly, not ones fetched after IHAVE
Out-of-subnet	our 2024 signal, used as an extra feature
41 h	until 60 % of peers are done (disconnect once all validators seen)
20×	fewer resources than fixed-length connections
91 %	classifier accuracy on held-out data
70 %	agreement with our 2024 mapping, high-confidence overlap

Doumanidis & Apostolaki, FC 2026, §4 · data Nov 2024 to Jan 2025

Where validators sit, late 2024

4,600	prefixes host all 1,063,660 validators
60 %	in 100 prefixes
37,000	validators in the largest prefix
> 80 %	in 20 ASes, out of 1,057 used
37 %	of validators in the US (31 % of nodes)
~90 %	of stake in 12 countries

Data Nov 2024 to Jan 2025 · Doumanidis & Apostolaki, FC 2026, §5

Which fix blunts which attack

	Subnet trick	Timing	StakeBleed	KnockBlock
All 64 subnets	●	○	○	○
Own subnet (DC)	●	○	○	○
No single-operator committees (DC)	●	○	○	○
No eager push, no out-of-subnet	●	●	○	○
Per-message anonymity (SPREAD)	○	●	○	○
Throwaway vote relay	●	●	○	○
Secret roles (SSLE, VRF, daily keys)	○	○	○	●
RPKI, diversity, failover	○	○	●	●

My reading of the sources on the previous slides; filled = blunts it, empty = does not